A Machine Learning-Based Model for Predicting Temperature Under the Effects of Climate Change

Mahmoud Y. Shams, Zahraa Tarek, Ahmed M. Elshewey, Maha Hany, Ashraf Darwish, and Aboul Ella Hassanien

1 Introduction

The geology, biology, and ecosystems of world have undergone significant, perhaps irreversible changes as a result of climate change [1]. Numerous environmental dangers to human health have resulted from these developments, including the worldwide spread of infectious illnesses, strain on food production systems, loss of biodiversity, and ozone layer depletion [2]. Low-lying island nations in the Pacific will be most affected by rising sea levels and erosion, which will damage houses and

M. Y. Shams (✉)
Faculty of Artificial Intelligence, Kafrelsheikh University, Kafr El-Shaikh 33516, Egypt
e-mail: mahmoud.yasin@ai.kfs.edu.eg

Z. Tarek
Faculty of Computers and Information, Mansoura University, Mansoura 35516, Egypt
e-mail: zahraatarek@mans.edu.eg

A. M. Elshewey
Faculty of Computers and Information, Department of Computer Science, Suez University, Suez, Egypt
e-mail: elshewy86@gmail.com

M. Hany
Scientific City for Research and Technology Applications, Alexandria, Egypt
e-mail: mhany2005@yahoo.com

A. Darwish
Faculty of Science, Helwan University, Cairo, Egypt
e-mail: ashraf.darwish.eg@ieee.org

A. E. Hassanien
Faculty of Computers and Artificial Intelligence, Cairo University, Giza, Egypt
e-mail: aboitcairo@cu.edu.eg

M. Y. Shams · Z. Tarek · A. M. Elshewey · M. Hany · A. Darwish · A. E. Hassanien
Scientific Research Group in Egypt, Giza, Egypt

A. E. Hassanien and A. Darwish (eds.), *The Power of Data: Driving Climate Change with Data Science and Artificial Intelligence Innovations*, Studies in Big Data 118,
https://doi.org/10.1007/978-3-031-22456-0_4

infrastructure and force people to evacuate. Rising ocean temperatures are linked to a rise and spread of illnesses in marine animals, and ingesting marine creatures puts people at risk of contracting these diseases directly [3, 4].

The process of monitoring climate change can be carried out through datasets that are collected to study its impact on the climate in general and provide recommendations for climate treatment. Artificial Intelligence (AI) can be a powerful tool in addressing some of humanity's greatest challenges. AI-driven systems help us to reduce the wasted energy amount in the home simply shutting off lighting and heating systems before we leave the home. Worldwide, these systems help combat drought by monitoring areas affected by desertification. Researchers are also looking about how data centers and AI computer systems themselves will affect the environment, such as finding a way to build systems and infrastructure that are more energy efficient. AI is just one tool in the complex process of analyzing the causes of climate change, but Its capacity to analyze massive volumes of data and find patterns enables us to comprehend the environment well. Machine learning enables AI systems to come up with solutions specific to those systems, rather than pre-programming them with a set of answers [5, 6].

Climate change is now a continuous event rather than something that will happen in the future. The reality of climate change has now been proved., and the changes create a greatest challenge for humanity today. So, climate change is the most burning issue now-a-days. It is evident that the global mean temperature has grown by 0.30–0.60 °C over last century. Due to a changing and disaster-prone climate, the global sea level has risen by 10–20 cm throughout the same time [7]. Egypt's climatic characteristics and weather pattern are undergoing some fundamental and significant changes. Temperatures are increasing, droughts and wildfires are becoming more common, rainfall patterns are changing, snow and glaciers are melting, and the globe sea level is increasing as a result of climate change. The emissions caused by human activity should be cut down on or completely stopped if we want to slow down climate change. Long-term changes in weather and temperature patterns are referred to as climate change. These changes might be caused by natural processes, such oscillations in the solar cycle. Since the 1800s, human activities—primarily the combustion of fossil fuels like oil, coal, and gas—have been the primary cause of climate change [8].

Forecasting is defined as the procedure of producing predictions based on early and current data. It is useful because it enables the creation of data-driven plans and the capacity to make educated judgments. Furthermore, it investigates how existing underlying currents imply prospective shifts in direction for enterprises, civilizations, or the entire planet. Prediction aims to provide future certainty. As a result, the primary goal of forecasting is to identify the whole range of possibilities rather than a small number of weak certainties. [9].

Air temperature prediction is the process of anticipating future temperature changes using a specific prediction model based on temperature time series data and other parameters. Temperature forecasting is used in weather forecasting, which can help to offer effective solutions to combat global warming. The ability to anticipate temperature variations is crucial for disaster warning, agricultural productivity, managing water resources, protecting the environment, and sustainable development. Recent years have seen an increase in the popularity of temperature forecasting on a worldwide scale [10].

The correlation between global warming and the increase in air temperature has lately caught experts' attention. The lives of people are eventually significantly impacted by climate change, which includes sea level rise, an increase in extreme occurrences, and global warming [11]. The atmosphere's state variable, air temperature, has an impact on both land surface and atmospheric processes. In order to safeguard people's lives and property, anticipating air temperature is a crucial component of weather prediction [12]. When the air temperature is outside of the acceptable range, people may experience possible health issues. Animals and plants may be harmed by abrupt temperature fluctuations. Due to the considerable impact that air temperature has on a number of industries, including agriculture, industry, and energy, an accurate prediction of air temperature is crucial. The precision of energy usage is improved by accurate air temperature forecasts [13]. Another important component in forecasting other meteorological variables is air temperature, including streamflow, solar radiation, and evapotranspiration. Finding a suitable method for air temperature forecasting is so essential and may help to lessen the effects of global warming and climate change. Furthermore, a strategy for human activities, economic growth, and energy policy, must take into account the precise forecast of air temperature [14–16]. The main contribution of this chapter is the studying the effect of climate change on the temperature by forecasting the global temperature using ML. We utilized standard dataset of climate change and perform preprocessing normalization for the selected features. Furthermore, we visualized the findings using both box and scatter plots. The evaluation results are performed using ML approaches; LR, RF KNN, DT, SVM, and CBR regressors.

The rest of the chapter is organized as follows. Section 2 briefly highlighted the current related work for predicting the global temperature. Section 3 presents the materials and method presented in this chapter. The proposed Cat Boost Regressor is presented in Sect. 4. The results and discussion are presented in Sect. 5. finally, the conclusion and perspectives are discussed in Sect. 6.

2 Related Work

Most research has focused on daily forecasting [17, 18], monthly [19] and annual mean temperatures [20, 21]. Hourly temperature forecasting has only received a small amount of research [22]. Traditional statistical methods including linear regression, cluster analysis, autoregressive integrated moving average (ARIMA), and grey prediction have been used to forecast air temperature up to this point. These models use statistical assessments of past data to determine the likelihood that a certain weather phenomenon will occur in the future. The mechanisms and variables driving variations in air temperature, on the other hand, are extremely intricate and nonlinear. Long time series of hourly or daily temperature cannot be successfully predicted using statistical methods because dynamic temperature changes are difficult to capture [9]. With the use of machine learning techniques like a support vector machine (SVM), the changing trend of air temperature has been anticipated [23], an artificial neural network (ANN) [24] and LSTM [25].

As an example, shallow learning approach, an SVM can estimate the maximum temperature for the following day across a time range of 2–10 days. An ANN, other shallow-learning technique, can accurately forecast changes in the daily average temperature trend [26]. A DBN and SAE, both deep learning techniques, are better at predicting temperature than a shallow neural network. A CNN (another deep learning approach) generates meteorological characteristics from convolution layers and sends them to a pooling layer, which selects and filters relevant information to limit the amount of input and prevent the CNN's gradient from disappearing [27]. As an extra deep learning approach, an RNN can anticipate time series of air temperature using neural units linked in a chain. A different deep-learning algorithm, LSTM, can anticipate short-term temperature correctly and effectively by gathering external input from hidden layers [28, 29].

To increase the accuracy of air temperature predictions, several deep-learning techniques have been included into models [30]. Convolutional recurrent neural network (CRNN) was created by combining a CNN with an RNN to identify the spatial and temporal correlations of the daily change in air temperature [31, 32].

Tran et al. [33] used standard LSTM, RNN, and multilayer ANN models. The highest air temperature in South Korea was predicted using hybrid models. The maximum air temperature for the next one to fifteen days was predicted using data from the previous seven days' worth of air temperature readings. The results demonstrated that the LSTM model outperformed the other models in terms of long-term air temperature predictions.

Tran and Lee make another attempt [34] utilized multilayer ANN models to forecast South Korea's maximum air temperature for one day. It was found that the ANN model generated the smallest error rate. For predicting air temperature, other research utilized deeper learning structures that were more complicated. As an example, Zhang et al. [31] used a convolutional recurrent neural network to predict the daily average air temperature for the next four days (CRNN). To train the CRNN, they used daily air temperature data collected across China between 1952 and 2018. Based on the historical air temperature data, the findings showed that their model could accurately forecast air temperature [26]. Cifuentes et al. [35] presented a deep learning approach to predict climate change temperature effect and they achieved minimum root mean square error 0.0017. As shown in Table 1, we summarize research that used neural network models to predict air temperature for a few minutes to many months in the future. Lin et al. [18] presents a temperature prediction based on multi-dimensional Empirical Decomposition Mode (EDM) ensemble and Radial Basis Function (RBF) with a minimum error rate in forecasting 7-day maximum temperature.

Table 1 The description of the current efforts to predict temperatures in presence of climate change

References	Input	Type of model	Output
Tran et al. [33]	Past daily maximum temperature (from 7 to 36)	Traditional ANN, RNN, LSTM	Daily maximum temperature for 1–15 days in advance
Tran and Lee [34]	Six previous daily maximum temperature	Traditional ANN	One day ahead maximum temperature
Zhang et al. [31]	Four past temperature data map series	Convolutional recurrent neural network (CRNN)	Four future temperature data map series
Cifuentes et al. [35]	Maximum, minimum, and average temperature, precipitation	Deep learning	Mean square error = 0.0017
Lin et al. [18]	Temperature, wind speed, humidity, and water depth are all factors to consider	Radial basis function neural network with multidimensional complementary ensemble empirical mode decomposition (MCEEMD) (RBFNN)	Forecast the daily maximum temperatures for the next seven days

3 Materials and Methods

The most significant environmental issue of the current era on a worldwide scale is climate change, which is brought on by global warming and is a consequence of rising greenhouse gas concentrations. Environmental issues brought on by climate change include shifting precipitation patterns, melting polar glaciers, and increasing sea levels [36]. Using a collection of climate change dataset, regression models were effective in predicting temperature in this chapter. This section provides a detailed discussion for the proposed model stages. The proposed model consists of data preprocessing for data normalization using Z normalization. After preparing and splitting the dataset, the regression models are applied to predict the temperature accurately. Different evaluation operators are used to evaluate the performance of the proposed prediction system. Results demonstrated that Cat boost regressor model achieved the best results with values of 0.003, 0.0036, 0.054, and 92.4% for MSE, MAE, RMSE, R^2, respectively. All steps of the proposed model are illustrated in Fig. 1.

Initially valid global dataset collected from the Climatic Research Unit at the University of East Anglia. Data was collected in a CSV file the data is normalized using Z normalization. Dataset is splitting to 70% for training, and 30% for testing. Multiple regression analyses were used interchangeably to predict temperature of the collected dataset related to climate change. The regression models e.g., Linear regression, Random regressor, K-nearest regressor, Decision tree regressor, Support

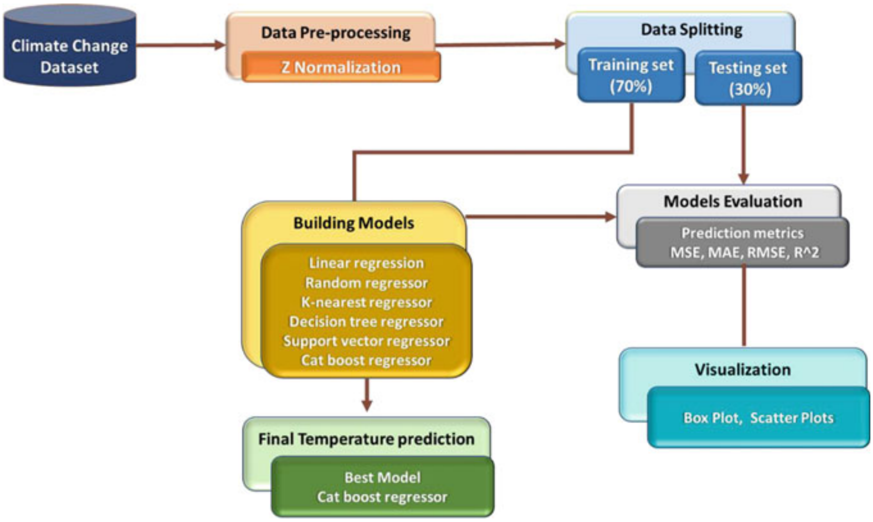

Fig. 1 The proposed temperature prediction model

vector regressor, have been trained with comparison to Cat boost regressor model for temperature prediction. Now, we'll discuss the regression models used in this chapter briefly.

3.1 Linear Regression

A well-known statistical technique, called linear regression, combines its input factors to predict the outcome variable in a straightforward and understandable manner [37]. One of linear regression's drawbacks is that it examines often a relationship between the mean of the output and input variables. Similar to how the mean does not adequately describe a single factor, linear regression does not provide a clear view of the relationships between variables [38]. Regression equation with numerous variables often look like as in Eq. (1):

$$Y = \beta + \beta_0 x_1 + \beta_1 x_2 + \beta_2 x_3 + \varepsilon. \tag{1}$$

where x_1, x_2, x_3 are the independent variables and Y is the predictor or target variable. β_0, β_1, β_2 are the coefficients and ε is the error term. Figure 2 presents the linear regression in machine learning [39].

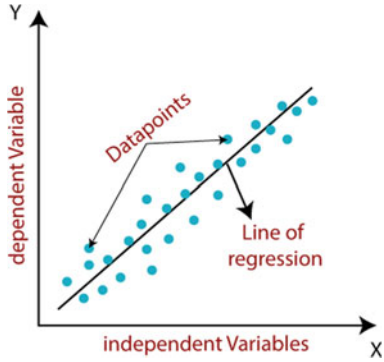

Fig. 2 Linear regression in machine learning

3.2 Decision Tree Regressor

One of the best regression techniques for handling high dimensional data is decision trees (DT). The decision trees are categories of non-parametric supervised learning techniques. The decision tree model learns "decision rules" derived from the properties of the current data to forecast the target variables. A piecewise constant approximation is similar to a decision tree regressor [40]. The decision tree divides recursively the feature space such that the samples with comparable target labels or values are grouped together.

Let Q_m with N_m samples represents the data at a node. Thus, the data is divided into $\mathbf{Q}_m^{left}$ (θ) and $\mathbf{Q}_m^{right}$ (θ) subsets for each split $\theta = (j,t_m)$ where j is a feature and t_m is a threshold [41], as shown in Eq. (2):

$$\mathbf{Q}_m^{left}(\theta) = \left\{(x, y)|x_j \leq t_m\right\}$$
$$\mathbf{Q}_m^{right}(\theta) = Q_m/\mathbf{Q}_m^{left}(\theta) \tag{2}$$

The variable in Eq. (3) that minimizes the impurity (θ^*) is:

$$\theta^* = \text{argmin}_\theta G\ (Q_m, \theta). \tag{3}$$

After the split's quality has been assessed, the parameters that reduce impurity denoted by θ^*.

There is recursively loop for subsets $\mathbf{Q}_m^{left}$ (θ) and $\mathbf{Q}_m^{right}$ (θ), until the maximum permitted depth is achieved with $N_m < \text{min}_{\text{samples}}$, $N_m = 1$.

3.3 Random Forest Regressor

The utilization of an ensemble of trees is the most effective technique to improve DT's prediction accuracy. A collection of regression trees that were built at random is known as Random Forest. Random Forest Regressor (RFR) is more appropriate for real-time applications in a variety of domains, such as species distribution modeling, language modeling, bioinformatics, and ecosystem modeling [42]. Different decision trees are trained using subsets of the dataset, and the final results are determined using the concept of majority voting (averaging). Therefore, among its competitors, ANN and XGBoost, random forest was shown to be the optimal technique through the approaches of bootstrap aggregation and replacement [43].

In RFR, a forest is formed from a collection of N trees $\{R_1(X), R_2(X), \ldots, R_N(X)\}$, where X is a p-dimensional input vector with values $x_1, x_2, \ldots, x_p$. The ensemble generates N outputs for each tree $\hat{Y}_n = R_1(X), \ldots, \hat{Y}_N = R_N(X)$, where $\hat{Y}_n$ is the nth tree output and $n = 1, \ldots, N$. The average of each tree's unique forecasts makes up the final result. The bootstrap sample is used to build the tree. After training is complete, fully formed trees are utilized to forecast results for samples that have not been seen or are unidentified. The learning error provided by the Mean Square Error (MSE) is used to measure the predictive performance of Random Forests. Equation (4) shows that $\hat{Y}(X_i)$ is the anticipated output, Y_i is the observed output correlating to a certain input sample and n is the total number of out-of-bag samples [42].

$$MSE \approx MSE^{OOB} = n^{-1} \sum_{i=1}^{n} \left(\hat{Y}(\mathrm{Xi}) - Y_i \right)^2. \tag{4}$$

3.4 K-Nearest Neighbor Regressor

The term "ML algorithms" refers to a type of algorithms that carry out computations implicitly through training rather than explicitly via programming. Most machine learning algorithms work by translating input characteristics to an output value (s). KNN are a class of ML algorithms that are reliable, easy to use, and cost little to run. The output value of a test point is specifically determined by KNN regressors as the interpolated value of the test point's closest neighbors. The number of neighbors to interpolate across, k, is used specified and the closest neighbors are determined by the k training points in the input feature space with the least Euclidean distance. Therefore, a weighted average of data points with comparable input properties may be used by KNN regressors to predict an output value [44]. Figure 3 illustrates the k-nearest neighbors regression [45].

3.5 Support Vector Regressor

SVR is a technique that may be used in regression situations when the support vector machine idea is being used. Due to its ability to address the overfitting issue, this

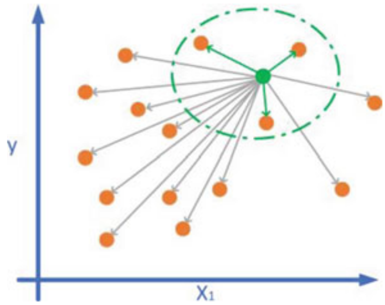

Fig. 3 k-nearest neighbors' regression

technique performs well in time series prediction and regression [46]. Equation (5) is used to represent the function in the linear case.

$$f(x) = (w, x) + b \, with \, w \in x, b \in R \tag{5}$$

whereas w denotes the slope, x presents the feature space, and b indicates the intercept. Equations (6) and (7) can be used in order to minimize the Euclidean value and make the function as flat as feasible.

$$minimum \, \frac{1}{2}\left\|w^2\right\| \tag{6}$$

depend on

$$\begin{cases} y_i - (w, x_i) - b \le \epsilon \\ (w, x_i) + b - y_i \le \epsilon \end{cases} \tag{7}$$

In contrast to SVM, which attempts to acquire a hyperplane by maximizing margins and has a limit of 1 by following $y_i(w.x_i - b) \ge 1$, , SVR strives for regression to achieve the value with the least error or minimal margin (2ε), such that the point is assumed to be inside the support hyperplane.

4 Cat Boost Regressor

Decision trees are used as the primary predictors in the majority of gradient boosting implementations. Decision trees are useful for numerical characteristics, but in reality, many datasets include categorical elements that are essential for prediction. A cutting-edge gradient boosting technique is the Cat Boost Regressor (CAT). It relates to more effective gradient boosting tree algorithmic framework implementation. A symmetrical decision tree algorithmic rule with categorical parameters, minimal variables, and superior accuracy is the foundation of this framework [47]. The CatBoost method, like Gradient Boosting and XGBoost, builds several binary decision trees every time it attempts to lower the error. The procedures for creating a tree in CatBoost are outlined in Algorithm 1 [48].

Algorithm 1: Constructing a tree in Cat Boost

input : $\{(x_p, y_p)\}_{p=1}^{n}$, M, L,α, $\{\sigma_s\}_{s=1}^{n}$,
Mod ← Calc_Grad(L, M, y);
d← random(1, n);
if Mod = Plain **then**
　R ← ($grad_r$(P) for P = 1..n);
if Mod = Ordered **then**
　R ← ($grad_{r,\sigma r(P)-1}$(P) **for** P = 1..n);
V ← vacuous tree;
foreach step from top to down **do**
　foreach candidate split c **do**
　　S_c ← add split c to S;
　　if Mod = Plain **then**
　　　Δ(p) ← avgerage($grad_r$(i) for i : $leaf_r$(i) = $leaf_r$(p)) for p = 1..n;
　　if Mod = Ordered **then**
　　　Δ(p) ← avgerage($grad_{r,\sigma r(p)-1}$(i) for i : $leaf_r$(i) = $leaf_r$(p), σ_r(i) < σ_r(p)) for p = 1..n;
　　loss(s_c) ← cos(Δ, R)
　V ← arg $\min_{Tc}$ (loss(S_c))
if Mod = Plain **then**
　$M_{\acute{b}}$(p) ← $M_{\acute{b}}$ (p) − α avgerage($grad_{\acute{b}}$ (i) for i : $leaf_{\acute{b}}$ (i) = $leaf_{\acute{b}}$ (p)) for $\acute{b}$ = 1..s, p = 1..n;
if Mod = Ordered **then**
　$M_{\acute{b},r}$ (p) ← $M_{\acute{b},r}$ (p) − α avgerage($grad_{\acute{b},r}$ (i) for i : $leaf_{\acute{b}}$ (i) = $leaf_{\acute{b}}$ (p), $\sigma_{\acute{r}}$ (i) ≤ r)
　for $\acute{b}$= 1..s, p= 1..n, r ≥ $\sigma_{\acute{b}}$ (p) − 1;
return V, M

When compared to several machine learning models, the benefits of gradient boosting algorithms include their superior prediction accuracy. These methods offer a great deal of versatility, allowing for the optimization of the function fit utilizing a variety of hyperparameter tuning choices or other loss functions. They don't need to preprocess the data since they can operate directly with the input. The main disadvantage of gradient boosting techniques is their high computational cost. Overfitting may result from these systems' elimination of mistakes, and the parameters have a significant impact on how they behave [49].

The algorithm's accuracy and its generalizability are enhanced by CAT. Numerous areas, including media popularity prediction, weather forecasting, biomass, and evapotranspiration, have effectively utilized this algorithm [50]. It is for this reason that the model is applied here to predict the temperature in this chapter.

5 Evaluation Results and Discussion

5.1 Dataset Description

The dataset is available at https://www.kaggle.com/datasets/econdata/climate-change. The dataset consists of 308 instances and 11 features. The features names are Year, Month, Temp, CO_2, N_2O, CH_4, CFC.11, CFC.12, Aerosols, TSI, MEI. The climate data from May 1983 to December 2008. The statistical analysis and determination of the applied features are shown in Table 2. The statistical includes count, mean, standard deviation (std.), minimum (min), maximum, 25, 50, and 75% of the features. Figure 4 demonstrates the boxplot of the applied features of the predicted temperatures. While the scatter plot that describe the relationship between the temperature and the year is shown in Fig. 5.

Figure 6 shows the relation between the temperature and the aerosols. Furthermore, Fig. 7, discussed the relation between the temperature and the months for

Table 2 Statistical calculation for the features

	Year	Month	MEI	CO_2	CH_4	N_2O	CFC-11	CFC-12	TSI	Aerosols	Temp.
Count	308	308	308	308	308	308	308	308	308	308	308
Mean	1995	6.5	0.27	363	1749	312	251	497	1366	0.01	0.25
Std.	7.4	3.4	0.93	12	46	5.2	20	57	0.39	0.02	0.17
Min	1983	1	– 1.6	340	1629	303	191	350	1365	0.001	– 0.28
25%	1989	4	– 0.39	353	1722	308	246	472	1365	0.002	0.12
50%	1996	7	0.23	361	1764	311	258	528	1365	0.005	0.24
75%	2002	10	0.83	373	1786	316	267	540	1366	0.012	0.40
Max	2008	12	3	388	1814	322	271	543	1367	0.14	0.73

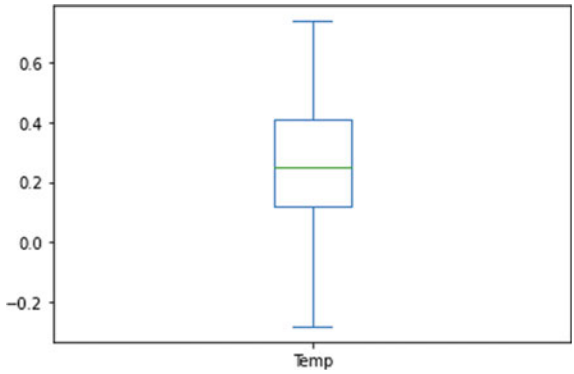

Fig. 4 Box plot for the applied temperature feature

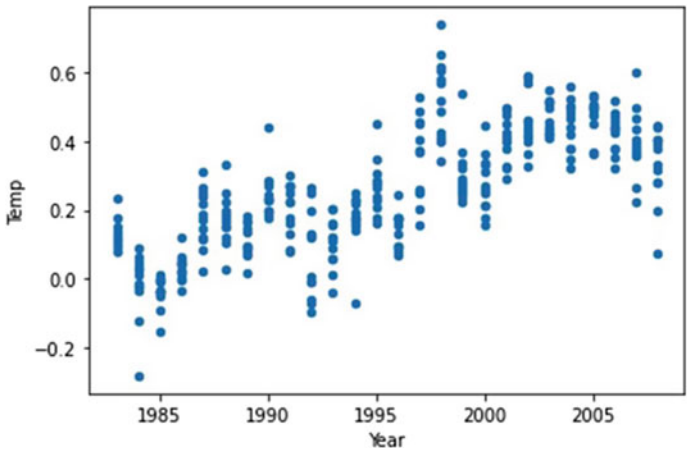

Fig. 5 Scatter plot between the temperature and years

one year. Figures 8, 9, 10, 11, 12, 13 and 14 describes the scatter plots relationship between the temperature and MEI, CO_2, CH_4, N_2O, CFC-11, CFC-12 and the TSI, respectively. The heatmap matrix that describes the relation and correlation between all the applied features (variables) of the dataset is shown in Fig. 15.

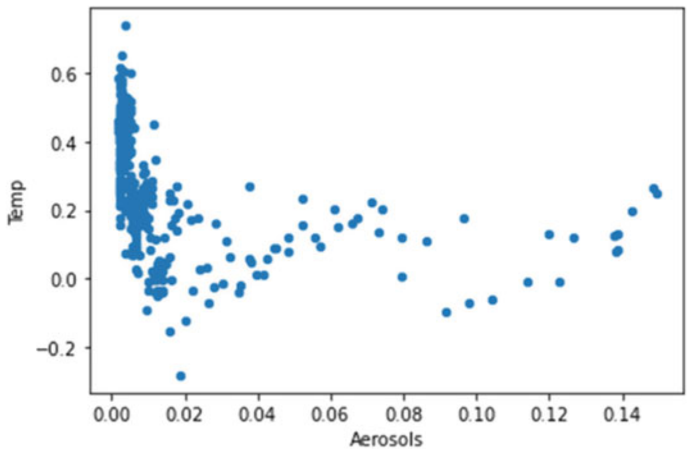

Fig. 6 Scatter plot between the temperature and aerosols

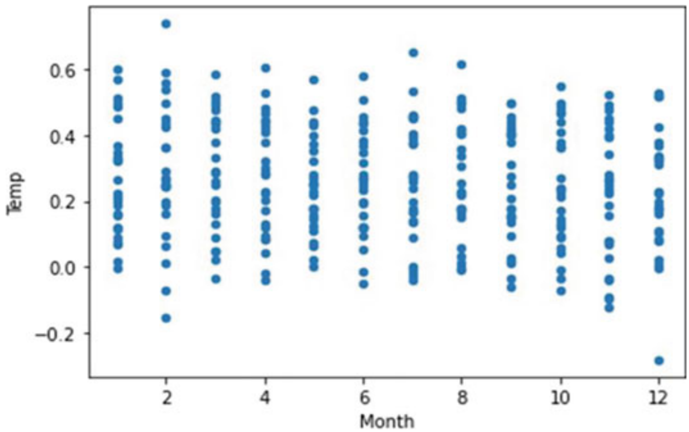

Fig. 7 Scatter plot between the temperature and month

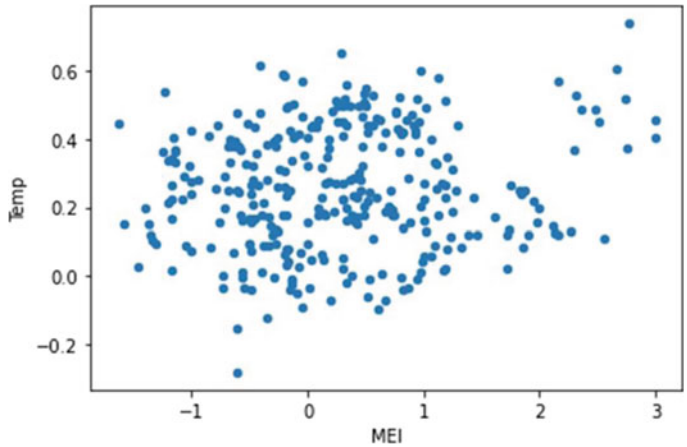

Fig. 8 Scatter plot between the temperature and MEI

5.2 *Evaluation Results*

The execution of the proposed regression ML (LR, RF KNN, DT, SVM, and CBR) models was assessed using the evaluation metrics, namely, Mean Square Error (MSE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and determination coefficient (R^2). MSE is computed using Eq. (8):

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(y_{actual_i} - y_{predicted_i}\right)^2 \tag{8}$$

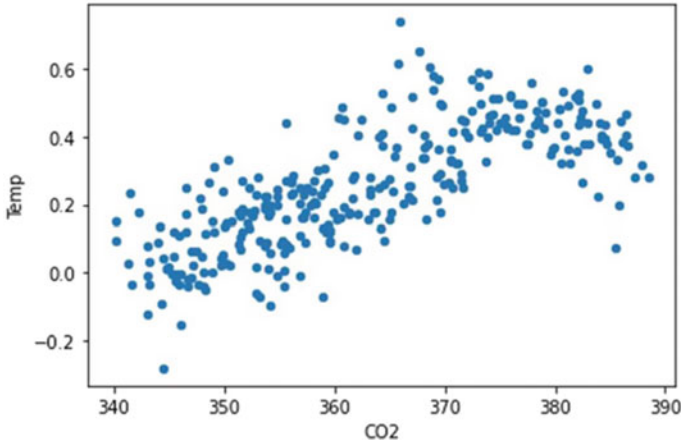

Fig. 9 Scatter plot between the temperature and CO_2

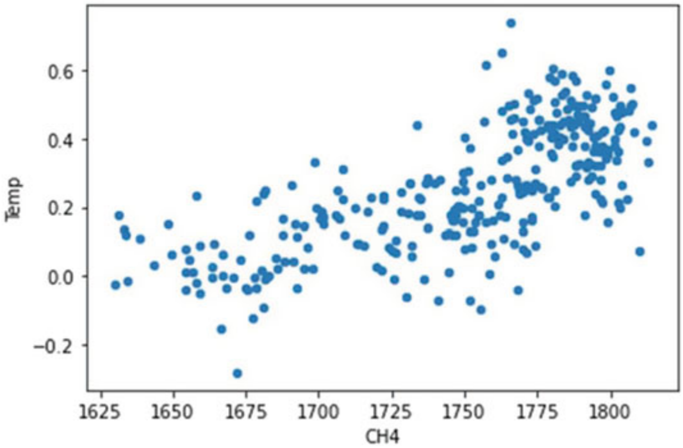

Fig. 10 Scatter plot between the temperature and CH_4

where N is the number of enrolled features and the y_{actual_i} is the actual values of the instance *i*. The predicted values of the instaces *i is denoted by* $y_{predicted_i}$. The RMSE, and the MAE are computed using Eqs. (9) and (10), respectively [51].

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_{actual_i} - y_{predicted_i}\right)^2} \tag{9}$$

$$MAE = \frac{1}{\text{N}}\sum_{i=1}^{N}\left|y_{actual_i} - y_{predicted_i}\right| \tag{10}$$

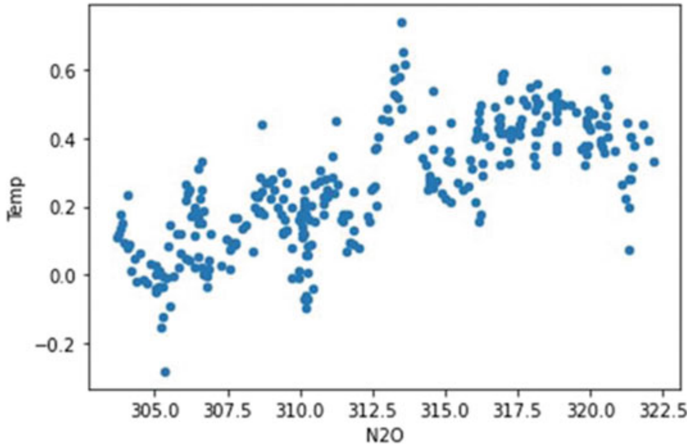

Fig. 11 Scatter plot between the temperature and N_2O

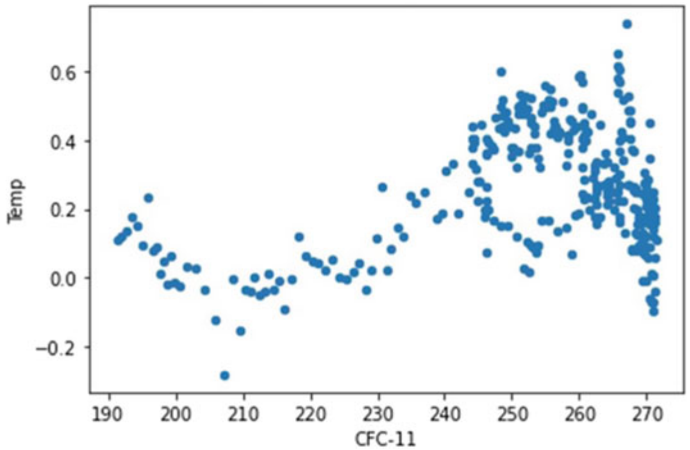

Fig. 12 Scatter plot between the temperature and CFC-11

The determination R^2 is computed using Eq. (11) as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_{actual_i} - y_{predicted_i}\right)^2}{\sum_{i=1}^{N}\left(y_{actual_i} - \overline{y}\right)^2} \tag{11}$$

where $\overline{y}$ is the mean value. Table 3 demonstrates the results obtained based on the mentioned evaluation metrices for the ML (LR, RF KNN, DT, SVM, and CBR) regressor models. Figure 16 shows the error rates (MSE, MAE and RMSE) for the proposed models. We noticed that the minimum error rates achieved when using

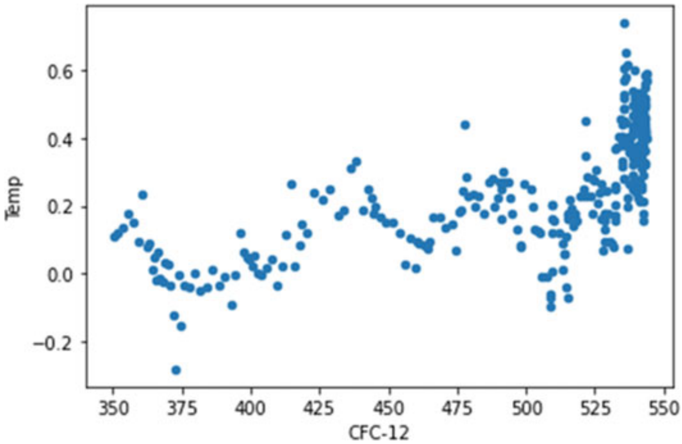

Fig. 13 Scatter plot between the temperature and CFC-12

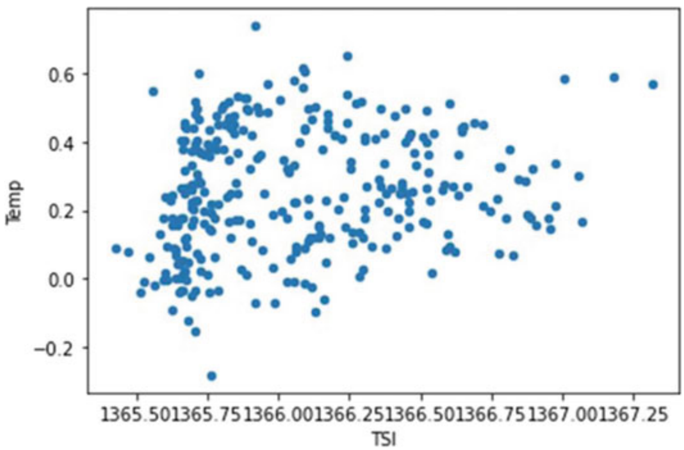

Fig. 14 Scatter plot between the temperature and TSI

Cat boot regressor. The results of determination coefficient (R^2) of the proposed ML models are shown in Fig. 17 indicated the superiority of the Cat boost regressor results (highlighted in bold) compared with other tradition ML regressor.

Fig. 15 The heatmap matrix for the features

Table 3 The results of the proposed ML models based on MSE, MAE, RMSE and R^2

Models	MSE	MAE	RMSE	R^2 (%)
Linear regression	0.009	0.0092	0.094	85.3
Random regressor	0.007	0.0075	0.787	88.9
K-nearest regressor	0.005	0.0058	0.070	90.6
Decision tree regressor	0.007	0.0071	0.787	88.4
Support vector regressor	0.004	0.0043	0.063	91.1
Cat boost regressor	**0.003**	**0.0036**	**0.054**	**92.4**

6 Conclusion and Perspectives

Climate change has a great effect on the surrounding environment. Currently the universe vulnerable great effect results from climate change. Especially the temperature which consider one of the important factors of climate change. This chapter present a prediction model based on ML regressors such as LR, RF KNN, DT, SVM, and CBR. Furthermore, we analysis the data using statistical analysis as well ae understanding the correlation between variables and the scatter plot relationship between the temperature and the other remaining features. These features are preprocessed and normalized using z-score normalization. Afterwards, it divided into 70% training

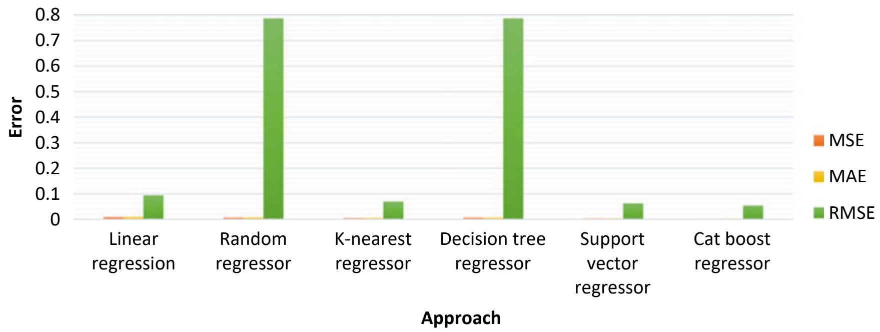

Fig. 16 The error values MSE, MAE and RMSE of the proposed ML models

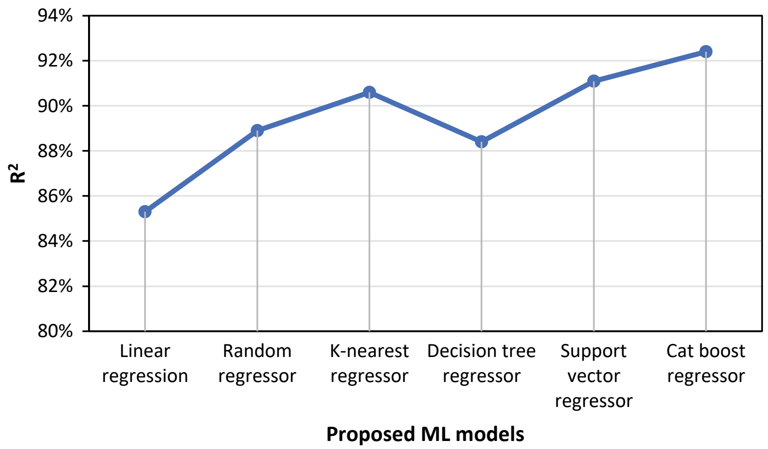

Fig. 17 The R^2 results compared with the proposed ML regressors

and the reminder 30% for testing. We used the prediction MSE, MAE, RMSE, and R^2 metrices. The results prove the ability of the proposed CBR model to predict the temperature with minimum error rates and high R^2 value compared with other ML models. In the future, we plan to utilize another climate change factor and recommend how can we tackle the change occurs. Moreover, we intended to study the effect of temperature on the agriculture fields especially the quality control of crops in the presence of climate change.

References

1. Steffen, W., Persson, Å., Deutsch, L., Zalasiewicz, J., & Williams, M., et al.: The Anthropocene: From global change to planetary stewardship. *AMBIO, 40*(7), 739 (2011). https://doi.org/10.1007/s13280-011-0185-x
2. Brierley, A. S., & Kingsford, M. J. (2009). Impacts of climate change on marine organisms and ecosystems. *Current Biology, 19*(14), R602–R614. https://doi.org/10.1016/j.cub.2009.05.046
3. Abraham, J. P., et al. (2013). A review of global ocean temperature observations: Implications for ocean heat content estimates and climate change. *Reviews of Geophysics, 51*(3), 450–483. https://doi.org/10.1002/rog.20022
4. Cheng, L., et al. (2021). Upper ocean temperatures hit record high in 2020. *Advances in Atmospheric Sciences, 38*(4), 523–530. https://doi.org/10.1007/s00376-021-0447-x
5. Fathi, S., Srinivasan, R. S., Kibert, C. J., Steiner, R. L., & Demirezen, E. (2020) AI-based campus energy use prediction for assessing the effects of climate change. *Sustainability, 12*(8), 8. https://doi.org/10.3390/su12083223
6. Chakraborty, D., Alam, A., Chaudhuri, S., Başağaoğlu, H., Sulbaran, T., & Langar, S. (2021). Scenario-based prediction of climate change impacts on building cooling energy consumption with explainable artificial intelligence. *Applied Energy, 291*, 116807. https://doi.org/10.1016/j.apenergy.2021.116807
7. Rahman, M. M., Opu, R. K., Sultana, A., & Riyad, M. R. A. (2020). *Climate change scenarios and analysis of temperature and rainfall intensity in Faridpur district*, Bangladesh
8. Hamed, M. M., Nashwan, M. S., & Shahid, S. (2022). Climatic zonation of Egypt based on high-resolution dataset using image clustering technique. *Progress in Earth and Planetary Science, 9*(1), 35. https://doi.org/10.1186/s40645-022-00494-3
9. Hou, J., Wang, Y., Zhou, J., & Tian, Q. (2022). Prediction of hourly air temperature based on CNN–LSTM. *Geomatics, Natural Hazards and Risk, 13*(1), 1962–1986. https://doi.org/10.1080/19475705.2022.2102942
10. Choi, B., Bergés, M., Bou-Zeid, E., & Pozzi, M. (2021). Short-term probabilistic forecasting of meso-scale near-surface urban temperature fields. *Environmental Modelling & Software, 145*, 105189. https://doi.org/10.1016/j.envsoft.2021.105189
11. Tajfar, E., Bateni, S. M., Lakshmi, V., & Ek, M. (2020). Estimation of surface heat fluxes via variational assimilation of land surface temperature, air temperature and specific humidity into a coupled land surface-atmospheric boundary layer model. *Journal of Hydrology, 583*, 124577. https://doi.org/10.1016/j.jhydrol.2020.124577
12. Tajfar, E., Bateni, S. M., Margulis, S. A., Gentine, P., & Auligne, T. (2020). Estimation of turbulent heat fluxes via assimilation of air temperature and specific humidity into an atmospheric boundary layer model. *Journal of Hydrometeorology, 21*(2), 205–225. https://doi.org/10.1175/JHM-D-19-0104.1
13. Valipour, M., Bateni, S. M., Gholami Sefidkouhi, M. A., Raeini-Sarjaz, M., & Singh, V. P. (2020). Complexity of forces driving trend of reference evapotranspiration and signals of climate change. *Atmosphere, 11*(10), 10. https://doi.org/10.3390/atmos11101081
14. Schulte, P. A., et al. (2016). Advancing the framework for considering the effects of climate change on worker safety and health. *Journal of Occupational and Environmental Hygiene, 13*(11), 847–865. https://doi.org/10.1080/15459624.2016.1179388
15. Marzo, A., et al. (2017). Daily global solar radiation estimation in desert areas using daily extreme temperatures and extraterrestrial radiation. *Renewable Energy, 113*, 303–311. https://doi.org/10.1016/j.renene.2017.01.061
16. Jovic, S., Nedeljkovic, B., Golubovic, Z., & Kostic, N. (2018). Evolutionary algorithm for reference evapotranspiration analysis. *Computers and Electronics in Agriculture, 150*, 1–4. https://doi.org/10.1016/j.compag.2018.04.003
17. Asha, J., Kumar, S. S., & Rishidas, S. (2021). Forecasting performance comparison of daily maximum temperature using ARMA based methods. *Journal of Physics Conference Series, 1921*(1), 012041. https://doi.org/10.1088/1742-6596/1921/1/012041

18. Lin, M.-L., Tsai, C. W., & Chen, C.-K. (2021). Daily maximum temperature forecasting in changing climate using a hybrid of multi-dimensional complementary ensemble empirical mode decomposition and radial basis function neural network. *Journal of Hydrology: Regional Studies, 38*, 100923. https://doi.org/10.1016/j.ejrh.2021.100923
19. Narasimha Murthy, K. V., Saravana, R., Kishore Kumar, G., & Vijaya Kumar, K. (2021). Modelling and forecasting for monthly surface air temperature patterns in India, 1951–2016: Structural time series approach. *Journal of Earth System Science, 130*(1), 21. https://doi.org/10.1007/s12040-020-01521-x
20. Liu, Z., et al. (2019). Balancing prediction accuracy and generalization ability: A hybrid framework for modelling the annual dynamics of satellite-derived land surface temperatures. *ISPRS Journal of Photogrammetry and Remote Sensing, 151*, 189–206. https://doi.org/10.1016/j.isprsjprs.2019.03.013
21. Johnson, Z. C., et al. (2020). Paired air-water annual temperature patterns reveal hydrogeological controls on stream thermal regimes at watershed to continental scales. *Journal of Hydrology, 587*, 124929. https://doi.org/10.1016/j.jhydrol.2020.124929
22. Carrión, D., et al. (2021). A 1-km hourly air-temperature model for 13 northeastern U.S. states using remotely sensed and ground-based measurements. *Environmental Research, 200*, 111477. https://doi.org/10.1016/j.envres.2021.111477
23. Gos, M., Krzyszczak, J., Baranowski, P., Murat, M., & Malinowska, I. (2020). Combined TBATS and SVM model of minimum and maximum air temperatures applied to wheat yield prediction at different locations in Europe. *Agricultural and Forest Meteorology, 281*, 107827. https://doi.org/10.1016/j.agrformet.2019.107827
24. Astsatryan, H., et al. (2021). Air temperature forecasting using artificial neural network for Ararat valley. *Earth Science Informatics, 14*(2), 711–722. https://doi.org/10.1007/s12145-021-00583-9
25. Bai, P., Liu, X., & Xie, J. (2021). Simulating runoff under changing climatic conditions: A comparison of the long short-term memory network with two conceptual hydrologic models. *Journal of Hydrology, 592*, 125779. https://doi.org/10.1016/j.jhydrol.2020.125779
26. Tran, T. T. K., Bateni, S. M., Ki, S. J., & Vosoughifar, H. (2021). A review of neural networks for air temperature forecasting. *Water, 13*(9), 9. https://doi.org/10.3390/w13091294
27. Bayatvarkeshi, M., et al. (2021). Modeling soil temperature using air temperature features in diverse climatic conditions with complementary machine learning models. *Computers and Electronics in Agriculture, 185*, 106158. https://doi.org/10.1016/j.compag.2021.106158
28. Mtibaa, F., Nguyen, K.-K., Azam, M., Papachristou, A., Venne, J.-S., & Cheriet, M. (2020). LSTM-based indoor air temperature prediction framework for HVAC systems in smart buildings. *Neural Computing and Applications, 32*(23), 17569–17585. https://doi.org/10.1007/s00521-020-04926-3
29. Sekertekin, A., Bilgili, M., Arslan, N., Yildirim, A., Celebi, K., & Ozbek, A. (2021). Short-term air temperature prediction by adaptive neuro-fuzzy inference system (ANFIS) and long short-term memory (LSTM) network. *Meteorology and Atmospheric Physics, 133*(3), 943–959. https://doi.org/10.1007/s00703-021-00791-4
30. Mohammadi, B., Mehdizadeh, S., Ahmadi, F., Lien, N. T. T., Linh, N. T. T., & Pham, Q. B. (2021). Developing hybrid time series and artificial intelligence models for estimating air temperatures. *Stochastic Environmental Research Risk Assessment, 35*(6), 1189–1204. https://doi.org/10.1007/s00477-020-01898-7
31. Zhang, Z., & Dong, Y. (2020). Temperature forecasting via convolutional recurrent neural networks based on time-series data. *Complexity, 2020*, e3536572. https://doi.org/10.1155/2020/3536572
32. Tabrizi, S. E., et al. (2021). Hourly road pavement surface temperature forecasting using deep learning models. *Journal of Hydrology, 603*, 126877. https://doi.org/10.1016/j.jhydrol.2021.126877
33. Thi Kieu Tran, T., Lee, T., Shin, J.-Y., Kim, J.-S., & Kamruzzaman, M. (2020). Deep learning-based maximum temperature forecasting assisted with meta-learning for hyperparameter optimization. *Atmosphere, 11*(5), 5. https://doi.org/10.3390/atmos11050487

34. Tran, T. T. K., Lee, T., & Kim, J.-S. (2020). Increasing neurons or deepening layers in forecasting maximum temperature time series? *Atmosphere, 11*(10), 10. https://doi.org/10.3390/atmos11101072
35. Cifuentes, J., Marulanda, G., Bello, A., & Reneses, J. (2020). Air temperature forecasting using machine learning techniques: A review. *Energies, 13*(16), 16. https://doi.org/10.3390/en13164215
36. Guo, L.-N., et al. (2021). Prediction of the effects of climate change on hydroelectric generation, electricity demand, and emissions of greenhouse gases under climatic scenarios and optimized ANN model. *Energy Reports, 7*, 5431–5445.
37. Liang, D., et al. (2022). Examining the utility of nonlinear machine learning approaches versus linear regression for predicting body image outcomes: The US body project I. *Body Image, 41*, 32–45.
38. Rath, S., Tripathy, A., & Tripathy, A. R. (2020). Prediction of new active cases of coronavirus disease (COVID-19) pandemic using multiple linear regression model. *Diabetes & Metabolic Syndrome: Clinical Research & Reviews, 14*(5), 1467–1474.
39. Javatpoint. (2021). Linear regression in machine learning. In *Javatpoint*. https://www.javatpoint.com/linear-regression-in-machine-learning. Accessed September 27, 2022.
40. An, Y., Wang, X., Qu, Z., Liao, T., & Nan, Z. (2018). Fiber Bragg grating temperature calibration based on BP neural network. *Optik, 172*, 753–759.
41. Dhanalakshmi, S., et al. (2022). Fiber Bragg grating sensor-based temperature monitoring of solar photovoltaic panels using machine learning algorithms. *Optical Fiber Technology, 69*, 102831.
42. Adusumilli, S., Bhatt, D., Wang, H., Devabhaktuni, V., & Bhattacharya, P. (2015). A novel hybrid approach utilizing principal component regression and random forest regression to bridge the period of GPS outages. *Neurocomputing, 166*, 185–192.
43. Mussumeci, E., & Coelho, F. C. (2020). Large-scale multivariate forecasting models for Dengue-LSTM versus random forest regression. *Spatial and Spatio-Temporal Epidemiology, 35*, 100372.
44. Durbin, M., Wonders, M. A., Flaska, M., & Lintereur, A. T. (2021). K-nearest neighbors regression for the discrimination of gamma rays and neutrons in organic scintillators. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, 987*, 164826.
45. Qaddoura, R., & Younes, M. B. (2022). Temporal prediction of traffic characteristics on real road scenarios in Amman. *Journal of Ambient Intelligence and Humanized Computing*, 1–16.
46. Kurniawan, R., Setiawan, I. N., Caraka, R. E., & Nasution, B. I. (2022). Using Harris hawk optimization towards support vector regression to ozone prediction. *Stochastic Environmental Research and Risk Assessment, 36*(2), 429–449.
47. Khan, P. W., Byun, Y.-C., Lee, S.-J., Kang, D.-H., Kang, J.-Y., & Park, H.-S. (2020). Machine learning-based approach to predict energy consumption of renewable and nonrenewable power sources. *Energies, 13*(18), 4870.
48. Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. *Advances in Neural Information Processing Systems, 31*.
49. Nguyen, V.-H., et al. (2021). Applying Bayesian optimization for machine learning models in predicting the surface roughness in single-point diamond turning polycarbonate. *Mathematical Problems in Engineering, 2021*.
50. Zhang, Y., Ma, J., Liang, S., Li, X., & Li, M. (2020). An evaluation of eight machine learning regression algorithms for forest aboveground biomass estimation from multiple satellite data products. *Remote Sensing, 12*(24), 4015.
51. Shams, M. Y., Tolba, A. S., & Sarhan, S. H. (2017). A vision system for multi-view face recognition. *International Journal of Circuits, Systems, and Signal Processing, 10*(1), 455–461. arXiv:1706.00510